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Engineering Thesis – Working Draft

Flight Dynamics Modelling of a Boeing 777-like Aircraft and Autopilot Design Using Nonlinear Model Predictive Control

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Abstract

This engineering thesis concerns the development of a nonlinear flight dynamics simulation environment for a Boeing 777-like transport aircraft and the design of an autopilot concept based on Nonlinear Model Predictive Control. The aircraft is treated as a large twin-engine wide-body transport configuration inspired by publicly available Boeing 777-300ER data, but the model is not intended to reproduce a certified manufacturer simulation model. The work therefore focuses on engineering consistency, transparent assumptions, and a modular software structure suitable for control design studies.

The thesis describes the adopted coordinate systems, state and input definitions, mass and inertia estimates, standard atmosphere model, first-cut aerodynamic model, propulsion assumptions, trimming requirements, and validation strategy. The implementation is organised as a MATLAB/Simulink-oriented project with reusable data modules, feature tests and reproducible data-processing tools. Particular attention is given to the separation between fixed aircraft properties, such as mass and inertia, and force quantities that depend on environmental assumptions such as gravitational acceleration.

The expected result of the project is a coherent simulation plant that can be trimmed in representative flight conditions and used as a basis for longitudinal and lateral-directional NMPC autopilot modes. The proposed controller architecture is designed to handle input limits, actuator rate limits, state constraints, and trajectory tracking objectives within a finite-horizon optimisation problem.

Keywords: flight dynamics, six-degree-of-freedom model, Boeing 777-like aircraft, nonlinear model predictive control, autopilot, MATLAB, Simulink.

Model Status and Data Notice

This document is a working engineering-thesis draft developed in parallel with the software implementation. The aircraft referred to as Boeing 777-like is a simplified engineering model of a large twin-engine transport aircraft. Its geometry, mass properties and propulsion assumptions are inspired by publicly available information about the Boeing 777 family, but the model is not an official Boeing model, does not use proprietary manufacturer data and must not be used for certification, operational planning or safety-critical analysis.

All numerical values that are not directly available from public reference material are treated as first-cut engineering estimates. These estimates are intended to support early integration of the flight dynamics model, trimming routines and control algorithms. They should be refined through validation, sensitivity studies and comparison with credible public performance data.

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Nomenclature

Symbol	Meaning
$\mathbf{r}_n$	position vector expressed in the navigation frame
$\mathbf{v}_b$	velocity vector expressed in the body frame
$\boldsymbol{\omega}_b$	angular velocity vector expressed in the body frame
ϕ, θ, ψ	Euler roll, pitch and yaw angles
p, q, r	body-axis roll, pitch and yaw rates
α	angle of attack
β	sideslip angle
V	true airspeed
q_∞	dynamic pressure
S	wing reference area
b	wing span
$\bar{c}$	mean aerodynamic chord
C_L, C_D, C_Y	lift, drag and side-force coefficients
C_l, C_m, C_n	rolling, pitching and yawing moment coefficients
δ_e	elevator deflection
δ_a	effective aileron deflection
δ_r	rudder deflection
T_1, T_2	left and right engine thrust
m	aircraft mass
$\mathbf{J}$	inertia matrix about the centre of gravity
g_0	standard gravitational acceleration used as the current project default

Abbreviation	Meaning
6-DOF	six-degree-of-freedom
CG	centre of gravity
ISA	International Standard Atmosphere
LNAV	lateral navigation

MAC	mean aerodynamic chord
MPC	model predictive control
NMPC	nonlinear model predictive control
NED	North-East-Down navigation frame
VNAV	vertical navigation

Chapter 1

Introduction

Automatic flight control systems are a fundamental part of modern transport aircraft. Their role extends beyond basic stabilisation: they reduce pilot workload, improve flight-path tracking, support energy management and help maintain operation within structural, aerodynamic and actuator constraints. The control problem is particularly demanding for large transport aircraft because of their high mass, large inertia, coupled longitudinal and lateral-directional dynamics, altitude-dependent performance and strict actuator limitations.

Classical autopilot architectures are often organised as hierarchical control systems. Inner loops stabilise angular rates or attitude variables, while outer loops generate commands for altitude, airspeed, heading or path tracking. This structure is effective and widely used, but it also separates tracking objectives from constraints. Nonlinear Model Predictive Control offers a different design philosophy: the nonlinear prediction model, performance objective and operational constraints are treated within one finite-horizon optimisation problem [1, 2].

The subject of this thesis is the development of a Boeing 777-like flight dynamics simulator and an NMPC-based autopilot concept. The phrase Boeing 777-like denotes an educational model of a large twin-engine wide-body aircraft whose dimensions and mass envelope are qualitatively close to the Boeing 777-300ER class. It does not denote an exact aircraft model. This distinction is important because accurate aerodynamic, engine and flight-control data for commercial aircraft are not publicly available in the detail required for a certified simulator.

1.1 Engineering Motivation

The project combines several areas of aerospace engineering: rigid-body flight mechanics, coordinate-frame conventions, aerodynamic modelling, propulsion modelling, numerical simulation, trimming and constrained control design. The value of the project lies in building the complete modelling chain. Instead of studying the controller in isolation,

the thesis develops a simulation plant, defines validation criteria and then uses the plant as the basis for autopilot synthesis.

The project also has a software-engineering motivation. A flight dynamics model becomes difficult to validate when geometry, mass data, atmosphere assumptions, aerodynamic coefficients and actuator limits are scattered through ad hoc files. For that reason, the implementation is organised into reusable MATLAB modules, with explicit units and feature-oriented checks. This makes the model easier to review and reduces the risk of hidden convention errors.

1.2 Problem Statement

The main engineering problem considered in this thesis can be stated as follows:

How can a coherent nonlinear flight dynamics model of a large transport aircraft be constructed from public and estimated data, and how can it be used as the basis for an NMPC autopilot that respects actuator, thrust and flight-envelope constraints?

This problem requires both modelling and control-design decisions. The simulation plant must be detailed enough to represent the dominant aircraft dynamics, but simple enough to be implemented, trimmed and validated within the scope of an engineering thesis. The NMPC prediction model must be computationally tractable while remaining dynamically consistent with the plant in the operating region of interest.

Chapter 2

Aim and Scope of the Thesis

2.1 Main Aim

The main aim of this thesis is to develop a nonlinear simulation environment for a Boeing 777-like transport aircraft and to design an autopilot concept based on Nonlinear Model Predictive Control. The simulator is intended to support analysis of six-degree-of-freedom aircraft motion, trimming in representative flight conditions and evaluation of constrained control algorithms.

2.2 Detailed Objectives

The detailed objectives of the work are:

1. to define coordinate frames, sign conventions, state vectors and input vectors used throughout the project,
2. to prepare a first-cut geometry, mass and inertia model of a Boeing 777-like aircraft,
3. to implement a simple International Standard Atmosphere model and fixed project constants,
4. to implement a simplified aerodynamic coefficient model suitable for early flight-dynamics integration,
5. to define a propulsion and actuator modelling approach,
6. to formulate the nonlinear six-degree-of-freedom equations of motion,
7. to prepare trimming and validation procedures for selected reference cases,
8. to design longitudinal and lateral-directional NMPC control structures,
9. to evaluate the model and controller using repeatable simulation scenarios.

2.3 Scope

The scope of the thesis includes rigid-body flight dynamics, simplified aerodynamics, simplified propulsion, actuator limits, standard atmosphere modelling, trimming, qualitative validation and NMPC autopilot design. The implementation target is MATLAB/Simulink, with MATLAB data modules used to define reusable aircraft and environment parameters.

The following topics are outside the scope of the present work:

- structural flexibility and aeroelastic effects,
- detailed hydraulic, electrical or avionics system models,
- certified flight-control laws,
- proprietary aerodynamic or engine data,
- high-fidelity pilot-display or flight-management-system simulation,
- certification-level validation.

2.4 Assumptions and Limitations

The aircraft is treated as a rigid body. The Earth is initially approximated as flat and non-rotating, and the navigation frame is a local North-East-Down frame. All internal calculations are performed in SI units. The current implementation uses fixed standard gravity $g_0 = 9.806\,65\text{ m/s}^2$ as a project-wide constant. Gravity-dependent force quantities are computed from mass and the selected gravity value rather than stored as immutable mass properties.

The aerodynamic model is intentionally simple. Its purpose is to provide physically reasonable signs, magnitudes and coupling terms for early simulation and control-design studies. The model is expected to be refined after trim tests and dynamic-response validation.

2.5 Success Criteria

The project will be considered successful if the following criteria are met:

1. the six-degree-of-freedom plant can be executed in simulation without numerical instability caused by modelling errors,
2. trim points can be obtained for several representative flight conditions,

3. open-loop dynamic responses are qualitatively consistent with expectations for a large transport aircraft,
4. the NMPC controller can stabilise and track selected flight variables,
5. actuator, thrust and selected state constraints are respected during simulation,
6. the obtained results are documented with plots, tables and engineering interpretation.

Chapter 3

Theoretical Background

3.1 Reference Frames in Flight Mechanics

Aircraft motion is described using several coordinate frames [3, 4]. The navigation frame used in this project is a local North-East-Down frame. Its axes point north, east and down, respectively. Altitude is therefore related to the down coordinate by $h = -D$.

The body frame is fixed to the aircraft. Its x_b axis points forward through the nose, its y_b axis points through the right wing and its z_b axis points downward. This convention is right-handed and is compatible with many aerospace simulation formulations. The wind frame is aligned with the air-relative velocity vector and is convenient for defining lift, drag, angle of attack and sideslip.

3.2 Six-Degree-of-Freedom Aircraft Motion

A six-degree-of-freedom aircraft model describes translational and rotational motion of a rigid body. The translational dynamics are obtained from Newton's second law expressed in a rotating body frame. The rotational dynamics are obtained from Euler's rigid-body equations. The state typically contains position, velocity, attitude and angular rates. In this project the state is extended by engine thrust and actuator states so that propulsion and control-surface dynamics are not assumed to be instantaneous.

3.3 Aerodynamic Coefficient Modelling

Aerodynamic forces and moments are commonly represented using nondimensional coefficients. Dynamic pressure is defined as

$$q_\infty = \frac{1}{2}\rho V^2, \tag{3.1}$$

where ρ is air density and V is true airspeed. Lift, drag and moments are then scaled using reference area, span and mean aerodynamic chord. For example,

$$L = q_\infty S C_L, \quad D = q_\infty S C_D, \quad M = q_\infty S \bar{c} C_m. \quad (3.2)$$

For an engineering first cut, the coefficients may be approximated by linear derivatives and simple nonlinear drag terms. Such a model is not sufficient for high-fidelity prediction near stall or in complex configurations, but it is adequate for early integration and control-design experiments when its validity range is clearly stated.

3.4 Propulsion and Actuator Dynamics

The propulsion model maps throttle commands to thrust. A realistic engine does not change thrust instantaneously, therefore the simulation model should include at least first-order spool dynamics. Similarly, control surfaces should include position limits, rate limits and possibly first-order actuator dynamics. These elements are important because NMPC relies on accurate constraint representation.

3.5 Nonlinear Model Predictive Control

Nonlinear Model Predictive Control solves an optimisation problem at each control update. The controller predicts future system behaviour over a finite horizon and selects an input sequence that minimises a cost function while satisfying constraints [1, 2]. Only the first input is applied to the plant, and the optimisation is repeated at the next sampling instant.

NMPC is attractive for aircraft autopilot design because it can explicitly include actuator limits, thrust limits, attitude limits, airspeed bounds, load-factor constraints and path-tracking objectives. Its main disadvantage is computational complexity, which motivates the use of reduced-order prediction models and carefully selected sampling times.

Chapter 4

Mathematical Model of the Aircraft

4.1 State Vector

The full nonlinear plant state is defined consistently with the project convention file as

$$\mathbf{x} = \begin{bmatrix} N & E & D & u & v & w & \phi & \theta & \psi & p & q & r & T_1 & T_2 & \delta_e & \delta_a & \delta_r \end{bmatrix}^T. \quad (4.1)$$

The variables N , E and D define position in the navigation frame. The variables u , v and w are body-axis velocity components. The Euler angles are roll ϕ , pitch θ and yaw ψ . The angular rates p , q and r are expressed in the body frame. The remaining states represent left and right engine thrust and actual control-surface deflections.

4.2 Command Input Vector

The command input vector is

$$\mathbf{u}_{cmd} = \begin{bmatrix} \delta_{e,cmd} & \delta_{a,cmd} & \delta_{r,cmd} & \tau_{1,cmd} & \tau_{2,cmd} \end{bmatrix}^T, \quad (4.2)$$

where $\delta_{e,cmd}$, $\delta_{a,cmd}$ and $\delta_{r,cmd}$ are commanded elevator, effective aileron and rudder deflections. The throttle commands $\tau_{1,cmd}$ and $\tau_{2,cmd}$ are nondimensional values in the range from zero to one.

4.3 Kinematics

The position kinematics are written as

$$\dot{\mathbf{r}}_n = \mathbf{C}_{nb} \mathbf{v}_b, \quad (4.3)$$

where $\mathbf{C}_{nb}$ transforms a vector from the body frame to the navigation frame. The Euler angle rates are related to body rates by

$$\begin{bmatrix} \dot{\phi} \\ \dot{\theta} \\ \dot{\psi} \end{bmatrix} = \begin{bmatrix} 1 & \sin \phi \tan \theta & \cos \phi \tan \theta \\ 0 & \cos \phi & -\sin \phi \\ 0 & \sin \phi / \cos \theta & \cos \phi / \cos \theta \end{bmatrix} \begin{bmatrix} p \\ q \\ r \end{bmatrix}. \quad (4.4)$$

This representation is simple and appropriate for normal aircraft attitudes, but it becomes singular as θ approaches $\pm 90^\circ$.

4.4 Translational Dynamics

The translational equations in the rotating body frame are

$$m\dot{\mathbf{v}}_b = \mathbf{F}_b - m\boldsymbol{\omega}_b \times \mathbf{v}_b, \quad (4.5)$$

where

$$\mathbf{v}_b = \begin{bmatrix} u & v & w \end{bmatrix}^T, \quad \boldsymbol{\omega}_b = \begin{bmatrix} p & q & r \end{bmatrix}^T. \quad (4.6)$$

The total body-frame force is decomposed as

$$\mathbf{F}_b = \mathbf{F}_{aero,b} + \mathbf{F}_{eng,b} + \mathbf{F}_{g,b}. \quad (4.7)$$

The gravitational force is computed using the current gravity assumption. At the present stage the project uses the fixed standard value g_0 , defined in the constants module.

4.5 Rotational Dynamics

The rotational equations are

$$\mathbf{J}\dot{\boldsymbol{\omega}}_b + \boldsymbol{\omega}_b \times (\mathbf{J}\boldsymbol{\omega}_b) = \mathbf{M}_b, \quad (4.8)$$

where $\mathbf{J}$ is the inertia matrix about the centre of gravity and $\mathbf{M}_b$ is the total body-frame moment. The first-cut inertia model uses

$$\mathbf{J} = \begin{bmatrix} I_{xx} & 0 & -I_{xz} \\ 0 & I_{yy} & 0 \\ -I_{xz} & 0 & I_{zz} \end{bmatrix}. \quad (4.9)$$

The current implementation sets $I_{xz} = 0$ to keep the early model simple. This assumption should be revisited after the plant can be trimmed and linearised.

4.6 Mass and Inertia Model

The mass module defines the mass envelope, nominal cruise mass, nominal approach mass and estimated inertia tensor. Because public certified inertia data for the aircraft are not available, the inertia tensor is estimated using effective radii of gyration based on public aircraft dimensions. This method is suitable for early integration but must be treated as an engineering approximation.

The implementation separates mass properties from gravity-dependent forces. The function `b777_mass_inertia.m` returns a structure containing mass and inertia values, while force quantities are obtained by evaluating a function handle:

```

1 mass = b777_mass_inertia();
2 forces = mass.forces(mass.standard_gravity_mps2);

```

Listing 4.1: Current gravity-dependent mass force interface.

This design allows the project to use fixed standard gravity at the current stage while preserving a clean interface for a future variable-gravity environment model.

4.7 Atmosphere Model

The atmosphere model is a simplified International Standard Atmosphere implementation up to 20 km. It returns temperature, pressure, density, speed of sound and viscosity. The same standard gravity constant used by the mass module is used in the hydrostatic pressure relations. This avoids hidden inconsistencies between atmosphere and mass-force calculations.

4.8 Aerodynamic Model

The aerodynamic model was built as a staged, source-oriented database rather than as a single analytical coefficient block. This choice was made because a Boeing 777-class aircraft cannot be represented credibly by one simplified polar while still retaining the traceability required for trimming, linearisation and NMPC design. The implemented interface is provided by `b777_aero_database.m`. It separates the plant-facing coefficient request from the source, status and confidence level of each coefficient group.

4.8.1 Development Objective and Data Strategy

The main difficulty is that detailed Boeing 777 aerodynamic tables are not public. The model is therefore deliberately a B777-like engineering model, not a certified Boeing model. The development strategy is to combine public transport-aircraft aerodynamic

data with semi-empirical derivative estimates and then accept data into the active plant only after validation gates have been passed.

The adopted source hierarchy is:

- B777-like geometry, reference area, span, mean aerodynamic chord and centre-of-gravity data from the project aircraft data modules [5, 6],
- NASA Common Research Model data for the clean-cruise longitudinal baseline [7, 8, 9],
- Digital DATCOM for missing static, damping and control derivative candidates [10, 11, 12],
- CRM-HL and DATCOM-ready high-lift increments for approach and landing development [13],
- OpenFOAM as a later verification tool for selected cases, not as the primary database source [14].

4.8.2 Development Pipeline

The construction path is deliberately staged. Geometry and mass define the reference quantities. NASA CRM data provide the active clean-cruise baseline. The analytical seed remains available as a complete fallback. Digital DATCOM Mach-grid, control-grid and derived gap-fill outputs are stored as candidate derivative data until validation gates allow promotion. CRM-HL/DATCOM-ready increments provide the first approach and landing configuration layer. The key rule is that source data, candidate data and active plant data are not treated as the same thing.

4.8.3 Clean-Cruise Baseline

At the current implementation level, the active clean-cruise longitudinal table is a Mach-alpha grid derived from NASA CRM NTF Test 197 TWICS-corrected force and moment data. The selected clean candidate is identified in the source files by `CONFIG=3`, `CONFTS=1` and `CONFT=0`. The table stores C_L , C_D and C_m over $M \approx 0.70$ to 0.87 and $\alpha = -2^\circ$ to 9° . Linear interpolation in Mach number and angle of attack is used only inside this committed range.

The CRM geometry is not interpreted as the Boeing 777 geometry. Instead, CRM is used as a public, well documented transport-aircraft aerodynamic baseline. The B777-like reference geometry is used later when the nondimensional coefficients are converted into forces and moments.

4.8.4 Residual Derivative and Control Layers

The clean CRM grid does not provide the complete coefficient set required by a six-degree-of-freedom plant. Residual effects are therefore added through static, damping and control derivative layers. In compact form, the assembled coefficient vector can be written as

$$\mathbf{C}_{aero} = \mathbf{C}_{clean}(M, \alpha) + \Delta\mathbf{C}_{\beta}(\beta) + \Delta\mathbf{C}_{rates}(\hat{p}, \hat{q}, \hat{r}) + \Delta\mathbf{C}_{ctrl}(\delta_e, \delta_a, \delta_r) + \Delta\mathbf{C}_{config}, \quad (4.10)$$

where $\mathbf{C}_{clean}$ contains the CRM clean longitudinal coefficients, $\Delta\mathbf{C}_{\beta}$ contains sideslip effects, $\Delta\mathbf{C}_{rates}$ contains damping effects, $\Delta\mathbf{C}_{ctrl}$ contains control-surface increments and $\Delta\mathbf{C}_{config}$ contains flap, slat and gear increments. The present active derivative values are still low-confidence engineering seeds, but their schema matches the intended replacement path through Digital DATCOM.

A reproducible Digital DATCOM package has been added before activating any DATCOM-derived values. Clean B777-like static and damping decks are run at $M = 0.30, 0.50, 0.60, 0.70$ and 0.80 , and the extracted derivatives are consolidated into a Mach-dependent candidate table. A companion control-grid workflow extracts elevator and aileron effectiveness candidates from **SYMFLP** and **ASYFLP** cases over the same Mach stations. Terms not printed directly by the current DATCOM outputs, including $C_{D\beta}$, C_{Y_r} , rudder derivatives and lateral control drag increments, are stored in a derived gap-fill candidate table. These gap-fill values are computed from the DATCOM candidate derivatives and the B777-like geometry, so they preserve active-interface coverage without being treated as direct DATCOM printouts.

4.8.5 Analytical Seed Model

The first-cut seed model provides lift, drag, side-force and moment coefficients as functions of angle of attack, sideslip, nondimensional angular rates and control-surface deflections. A representative longitudinal form is

$$C_L = C_{L0} + C_{L\alpha}\alpha + C_{Lq}\hat{q} + C_{L\delta_e}\delta_e, \quad (4.11)$$

$$C_m = C_{m0} + C_{m\alpha}\alpha + C_{mq}\hat{q} + C_{m\delta_e}\delta_e, \quad (4.12)$$

with a drag model of the form

$$C_D = C_{D0} + kC_L^2 + C_{D\beta}\beta^2 + C_{D\delta_e}\delta_e^2 + C_{D\delta_a}\delta_a^2 + C_{D\delta_r}\delta_r^2. \quad (4.13)$$

The coefficient signs follow the body-frame and control-surface conventions defined in the project.

The seed model has two roles. First, it provides a simple complete coefficient set for

software integration. Second, it defines the fallback behaviour outside the validated CRM grid. Whenever the database uses fallback values, the source metadata indicate that the coefficient group is not coming from validated table data.

4.8.6 Configuration and High-Lift Layer

The database layer also stores configuration metadata for clean, approach and landing cases. The current clean configuration uses the CRM grid only inside its local validity range. Inside that range, the CRM grid supplies the clean longitudinal baseline and the derivative table supplies residual lateral-directional, damping and control effects.

The approach and landing configurations use a separate CRM-HL/DATCOM-ready high-lift seed table with explicit flap, slat, gear, lift-increment, drag-increment, pitching-moment-increment and $C_{L,\max}$ metadata. The $C_{L,\max}$ value is currently exposed for trim and validation checks, not as a complete stall model. These values are sufficient for software integration and initial trim testing, but they are not intended to represent certified aircraft data.

4.8.7 Candidate Promotion Policy

Static, control and damping derivatives are exposed through dedicated tables so that DATCOM or validation-derived estimates can later replace individual groups without changing the plant interface. The current DATCOM Mach-grid, control-grid and derived gap-fill files are therefore treated as review artifacts rather than active aerodynamic tables.

The DATCOM candidates are controlled by automatic promotion gates. The gates check source identifiers, case traceability, run quality, Mach-grid consistency, coefficient signs, confidence thresholds, control-derivative coverage and active-interface coverage. Clean-cruise longitudinal trim validation and restricted longitudinal modal validation have passed for three DATCOM-candidate cases, but the candidate derivatives remain inactive until the warning review is complete. This prevents a partially reviewed candidate table from silently entering the nonlinear plant. The final aerodynamic database should be treated as acceptable for NMPC studies only after trim, modal and performance validation gates have been passed.

4.8.8 Conversion to Forces and Moments

The plant-facing aerodynamic load model is implemented by `b777_aero_forces_moments.m`. It evaluates the coefficient database, computes dynamic pressure, transforms the wind-axis force vector into the body frame and scales the moment coefficients by

the project reference span and mean aerodynamic chord:

$$\mathbf{F}_w = q_\infty S \begin{bmatrix} -C_D & C_Y & -C_L \end{bmatrix}^T, \quad (4.14)$$

$$\mathbf{F}_b = \mathbf{C}_{bw}(\alpha, \beta) \mathbf{F}_w, \quad (4.15)$$

$$\mathbf{M}_{ref,b} = q_\infty S \begin{bmatrix} bC_l & \bar{c}C_m & bC_n \end{bmatrix}^T. \quad (4.16)$$

If the aerodynamic reference point is not coincident with the current centre of gravity, the moment is shifted using

$$\mathbf{M}_{cg,b} = \mathbf{M}_{ref,b} + \mathbf{r}_{ref/cg,b} \times \mathbf{F}_b. \quad (4.17)$$

This step makes the aerodynamic database usable by the nonlinear plant while keeping the moment-reference policy explicit.

4.9 Actuator and Engine Model

The plant applies command signals through actuator and engine dynamics rather than feeding commanded values directly into the force and moment equations. The current control-surface actuator model covers elevator, effective aileron and rudder deflections. Each commanded deflection is first limited to the available surface range:

$$\delta_{cmd,lim} = \text{sat}_{\delta_{min}, \delta_{max}}(\delta_{cmd}). \quad (4.18)$$

The actual surface state then follows a rate-limited first-order lag,

$$\dot{\delta} = \text{sat}_{\dot{\delta}_{min}, \dot{\delta}_{max}} \left(\frac{\delta_{cmd,lim} - \delta}{\tau_\delta} \right), \quad (4.19)$$

with position limits

$$\delta_{min} \leq \delta \leq \delta_{max}. \quad (4.20)$$

If a surface is already at a position limit, the rate command is prevented from driving it farther outside the admissible range. The first implementation uses engineering estimates of the limits and time constants: elevator $\pm 25^\circ$, effective aileron $\pm 20^\circ$ and rudder $\pm 30^\circ$. The actuator output passed to the aerodynamic model is the current actual deflection, not the commanded value. This separation is required for NMPC because the controller must respect both surface deflection limits and surface rate limits.

The current propulsion model represents two under-wing turbofan engines. It is not a certified GE90 engine deck; it is an engineering seed based on public maximum static thrust information and the B777-like engine locations defined in the geometry module

[15]. The maximum available thrust per engine is modelled as

$$T_{\max}(h, M) = T_{SL} \left(\frac{\rho(h)}{\rho_0} \right)^{a_T} \max(f_{M,\min}, 1 - k_M M), \quad (4.21)$$

where T_{SL} is the sea-level static thrust reference, $\rho(h)$ is obtained from the ISA atmosphere model and M is Mach number. The coefficients a_T , k_M and $f_{M,\min}$ are first-cut engineering parameters and are kept explicit so that later performance tuning can replace them without changing the plant interface.

Throttle commands are mapped from idle to maximum available thrust:

$$T_{idle}(h, M) = f_{idle} T_{\max}(h, M), \quad (4.22)$$

$$T_{cmd,i} = T_{idle}(h, M) + \tau_i (T_{\max}(h, M) - T_{idle}(h, M)), \quad 0 \leq \tau_i \leq 1. \quad (4.23)$$

The actual thrust states are then propagated with a first-order spool model,

$$\dot{T}_i = \frac{T_{cmd,i} - T_i}{\tau_T}, \quad (4.24)$$

where $i \in \{1, 2\}$ denotes the left and right engine. These dynamics are essential for NMPC because they define the achievable rate of change of the longitudinal force and of asymmetric-thrust yawing moment.

The plant-facing engine load adapter converts the actual thrust states into body-frame forces and moments. With the current thrust direction aligned with the positive body x axis,

$$\mathbf{F}_{eng,b} = \sum_{i=1}^2 T_i \mathbf{d}_{T,b}, \quad (4.25)$$

$$\mathbf{M}_{eng,b} = \sum_{i=1}^2 \left(\mathbf{r}_{i/cg,b} \times T_i \mathbf{d}_{T,b} \right). \quad (4.26)$$

The engine position vectors are taken from the B777-like geometry module. This means that symmetric thrust produces a longitudinal force and a pitching moment if the thrust line is vertically offset from the centre of gravity, while differential thrust produces a yawing moment because the engines are located outboard of the aircraft centreline. Fuel flow, engine gyroscopic moments, reverse thrust, nacelle drag and detailed spool scheduling are left for a later propulsion-model refinement.

Chapter 5

Implementation

5.1 Software Organisation

The implementation is organised as a modular MATLAB/Simulink workspace. The `data` directory contains reusable aircraft, atmosphere, aerodynamic, propulsion and actuator data. The `plant` directory contains adapters that convert those data into forces, moments and state derivatives. The `tests` directory contains feature-oriented MATLAB checks, while `tools` stores scripts used to reproduce DATCOM extraction and validation artifacts.

The main implementation rule is separation of concerns. Fixed aircraft properties, environment models, aerodynamic data, actuator dynamics and plant force build-up are kept in separate modules. This keeps each assumption traceable and allows individual layers to be replaced without changing the full plant interface.

5.2 Implemented Modules

The current data layer contains:

- geometry, mass, inertia, constants and ISA atmosphere modules,
- a staged aerodynamic database with CRM clean-cruise data, DATCOM-ready derivative layers and high-lift increments,
- a B777-like propulsion seed with thrust lapse and spool dynamics,
- a control-surface actuator seed with position and rate limits,
- reference cases, configurations and modelling conventions.

The current plant-facing adapters are:

- `b777_aero_forces_moments.m`, which converts aerodynamic coefficients into body-axis forces and moments,
- `b777_engine_forces_moments.m`, which converts left and right engine thrust into body-axis propulsion loads,
- `b777_actuator_dynamics.m`, which maps commanded control-surface deflections to actual surface-state derivatives.

5.3 Aerodynamic Data Processing

The aerodynamic database is built from versioned source files and reproducible helper scripts. NASA CRM data provide the active clean-cruise longitudinal baseline. Digital DATCOM outputs are parsed into candidate derivative tables for static, damping and control effects. Derived gap-fill candidates cover interface terms that are not printed directly by the current DATCOM workflow. Trim, modal and promotion-gate reports are stored as validation evidence before candidate data are promoted.

Generated DATCOM run folders are treated as reproducible workspace outputs and are not stored as source data. The committed repository retains the input decks, candidate CSV files, reports and validation artifacts needed to audit the modelling chain.

5.4 Feature Tests

Feature tests check module interfaces and modelling conventions rather than certifying aircraft fidelity. They currently verify:

- project structure and MATLAB path assumptions,
- state, input, unit and sign conventions,
- geometry, mass, inertia, atmosphere and reference-case consistency,
- aerodynamic source inventory, coefficient behaviour and load signs,
- engine thrust mapping, thrust lapse, spool derivatives and asymmetric thrust moments,
- actuator position limits, rate limits and command-to-state mapping.

These tests are intended to catch convention errors early. Physical validation is handled separately through trim, modal and later simulation studies.

5.5 Current Status

At the current stage the project contains the reusable data layer, aerodynamic load adapter, propulsion load adapter and control-surface actuator adapter. The next implementation step is to assemble the total force and moment build-up and integrate the nonlinear six-degree-of-freedom equations in MATLAB/Simulink.

Chapter 6

Trimming and Validation

6.1 Role of Trimming

Trimming is a central step in flight dynamics modelling. A trim condition is a steady or quasi-steady operating point in which selected accelerations and tracking errors are zero or match prescribed values. For level cruise, for example, the aircraft should maintain altitude and airspeed with approximately zero angular accelerations and zero flight-path angle.

Without trim points it is difficult to evaluate stability, initialise simulations, linearise the model or design an autopilot. In this project trimming is therefore treated as part of the model-development workflow rather than as a post-processing task. A candidate aerodynamic database is not considered ready for further use merely because its coefficients have plausible signs; it must also allow the aircraft to reach physically reasonable equilibrium conditions.

6.2 Trim Formulation

A trim problem can be expressed as a constrained nonlinear optimisation problem:

$$\min_{\mathbf{x}, \mathbf{u}} \|\mathbf{f}_{trim}(\mathbf{x}, \mathbf{u}, \mathbf{p})\|^2, \quad (6.1)$$

where $\mathbf{x}$ is the state vector, $\mathbf{u}$ is the input vector and $\mathbf{p}$ contains prescribed flight-condition parameters such as altitude, speed, mass and configuration. The residual vector may include translational accelerations, angular accelerations, flight-path-angle error, sideslip error and actuator equilibrium errors.

6.3 Implemented Longitudinal Trim Check

At the current stage, a restricted clean-cruise longitudinal trim check has been implemented for the DATCOM candidate aerodynamic path. This check does not yet replace the future full six-degree-of-freedom trim solver. Its purpose is narrower: it verifies that the CRM clean baseline combined with the DATCOM elevator candidates can produce realistic cruise equilibrium points before the derivative candidates are considered for promotion.

For a prescribed altitude h , Mach number M , mass m and flight-path angle γ , the speed and dynamic pressure are computed from the ISA atmosphere:

$$V = Ma(h), \quad (6.2)$$

$$q_\infty = \frac{1}{2}\rho(h)V^2. \quad (6.3)$$

For the current level-cruise cases, the attitude relation is approximated by

$$\theta = \alpha + \gamma. \quad (6.4)$$

The elevator deflection is obtained from the zero pitching-moment condition:

$$C_m(M, \alpha, \delta_e) = C_{m, \text{clean}}(M, \alpha) + C_{m_{\delta_e}}(M)\delta_e = 0, \quad (6.5)$$

which gives

$$\delta_e = -\frac{C_{m, \text{clean}}(M, \alpha)}{C_{m_{\delta_e}}(M)}. \quad (6.6)$$

The remaining scalar unknown is the angle of attack. It is solved from the vertical body-axis force balance:

$$r_z(\alpha) = q_\infty S (-C_D \sin \alpha - C_L \cos \alpha) + mg \cos \theta = 0. \quad (6.7)$$

After α and δ_e have been found, the required thrust is computed from the axial balance:

$$T = -q_\infty S (-C_D \cos \alpha + C_L \sin \alpha) + mg \sin \theta. \quad (6.8)$$

In this validation step, thrust is treated as a static-equivalent scalar along the body x axis. Engine pitching moments, actuator states and lateral-directional equilibrium are intentionally outside the scope of this first trim gate.

The implemented script `tools/run_trim_validation.py` evaluates three clean-cruise cases at 10 668 m and 250 000 kg. The case table is written to `data/aerodynamics/validation/trim_validation_cases.csv`, while the summary report is written to `docs/trim_validation_report.md`. The current results are listed in Table 6.1.

Table 6.1: Current clean-cruise longitudinal trim-validation results for the DATCOM candidate path.

Case	Mach	α [deg]	δ_e [deg]	C_L	T [kN]
clean_cruise_m070	0.70010	5.5815	-0.5315	0.6822	143.41
clean_cruise_m075	0.75015	4.4035	0.1550	0.5952	130.21
clean_cruise_m080	0.80000	3.4957	0.9323	0.5238	126.97

The acceptance criteria require the angle of attack to remain inside the clean CRM range and below 8° , the elevator trim to remain within $\pm 5^\circ$, the static-equivalent thrust to remain positive and below 30% of the public GE90-115B static thrust estimate, the residual forces to be numerically negligible, the pitching-moment coefficient residual to be near zero and the margin to the current clean $C_{L,\max}$ estimate to remain at least 0.40. All three cases pass these checks, so the validation report records the marker

$$\text{DATCOM_PROMOTION_TRIM_VALIDATED}=\text{true}. \quad (6.9)$$

This marker does not by itself activate the DATCOM candidate derivatives. It only removes the trim-validation blocker from the promotion gate; the modal gate and DATCOM warning review are evaluated separately.

6.4 Implemented Longitudinal Modal Check

The next validation layer uses the same three clean-cruise trim cases to check the dominant longitudinal modes of the DATCOM candidate path. The implemented script `tools/run_longitudinal_modal_validation.py` forms a restricted longitudinal state vector

$$\mathbf{x}_{lon} = \begin{bmatrix} u & w & q & \theta \end{bmatrix}^T. \quad (6.10)$$

The elevator and thrust values are held at their trim values. The aerodynamic model uses the CRM clean C_L , C_D and C_m table, DATCOM candidate C_{L_q} and C_{m_q} derivatives, and the DATCOM elevator effectiveness candidates. The resulting nonlinear residual is linearised numerically around each trim point:

$$\Delta \dot{\mathbf{x}}_{lon} = \mathbf{A}_{lon} \Delta \mathbf{x}_{lon}. \quad (6.11)$$

The finite-difference scheme uses central differences inside the data range and one-sided differences when a trim point lies on a committed Mach-grid boundary. This avoids extrapolating the CRM or DATCOM candidate data.

The current modal-validation results are listed in Table 6.2. The slower pair is classified as phugoid and the faster pair as short-period.

Table 6.2: Current restricted longitudinal modal-validation results for the DATCOM candidate path.

Case	Mode	$\Re(\lambda)$ [1/s]	ω_n [rad/s]	ζ	Period [s]
clean_cruise_m070	phugoid	-0.001215	0.063889	0.01902	98.36
clean_cruise_m070	short-period	-0.363103	0.652714	0.55630	11.58
clean_cruise_m075	phugoid	-0.001819	0.066811	0.02722	94.08
clean_cruise_m075	short-period	-0.425626	0.869361	0.48959	8.29
clean_cruise_m080	phugoid	-0.001723	0.064624	0.02666	97.26
clean_cruise_m080	short-period	-0.479470	1.016779	0.47156	7.01

All three cases produce two stable oscillatory mode pairs. The phugoid natural frequency and period remain in a low-frequency, lightly damped range, while the short-period mode remains faster and more strongly damped. Therefore the validation report records the marker

$$\text{DATCOM_PROMOTION_MODAL_VALIDATED}=\text{true}. \quad (6.12)$$

This result is still limited to the restricted longitudinal model. It does not validate lateral-directional modes, actuator dynamics, engine pitching moments or a fully coupled six-degree-of-freedom linearisation.

6.5 Remaining Validation Work

The current checks validate only a restricted clean-cruise longitudinal path. The next validation step is a full-plant trim and linearisation around selected reference cases:

$$\Delta \dot{\mathbf{x}} = \mathbf{A} \Delta \mathbf{x} + \mathbf{B} \Delta \mathbf{u}. \quad (6.13)$$

The resulting models should add lateral-directional and coupled six-degree-of-freedom modal checks before the aircraft model is used as the final NMPC validation plant.

Chapter 7

NMPC Autopilot Design

7.1 Autopilot Architecture

The autopilot is designed as a layered system. The guidance layer generates references such as altitude, airspeed, heading, flight-path angle or path error. The NMPC layer computes control commands that minimise tracking errors while respecting constraints. The actuator and propulsion layer maps commands to actual elevator, aileron, rudder and thrust states.

7.2 Optimisation Problem

At each control update the NMPC controller solves a finite-horizon optimisation problem. A representative cost function is

$$J = \sum_{k=0}^{N_p} \|\mathbf{y}_k - \mathbf{y}_{ref,k}\|_Q^2 + \sum_{k=0}^{N_c-1} \|\mathbf{u}_k - \mathbf{u}_{ref,k}\|_R^2 + \sum_{k=0}^{N_c-1} \|\Delta \mathbf{u}_k\|_S^2. \quad (7.1)$$

The terms in the cost function penalise tracking errors, absolute control usage and control increments. Additional soft-constraint penalties may be added for flight-envelope variables when strict feasibility is difficult to guarantee.

7.3 Constraints

The most important constraints are:

- elevator, aileron and rudder deflection limits,
- control-surface rate limits,
- thrust and throttle limits,

- bank-angle and pitch-angle limits,
- angle-of-attack and sideslip limits,
- airspeed bounds,
- vertical-speed or flight-path-angle bounds,
- load-factor bounds.

The explicit treatment of these limits is the main reason for selecting NMPC rather than a purely classical autopilot design for this project.

7.4 Reduced Prediction Models

The first implementation will use separate reduced prediction models for longitudinal and lateral-directional control. The longitudinal model will include airspeed, angle of attack, pitch rate, pitch angle, altitude and equivalent thrust. The lateral-directional model will include sideslip, roll rate, yaw rate, bank angle, heading angle and lateral path error. This separation reduces computational complexity and allows each channel to be validated independently before any integrated NMPC formulation is attempted.

7.5 Planned Autopilot Modes

The planned modes include:

- altitude hold,
- speed hold,
- vertical-speed or flight-path-angle control,
- heading select,
- simplified lateral navigation tracking,
- simplified vertical navigation tracking,
- localiser and glideslope capture in a simplified approach scenario.

Chapter 8

Simulation Studies

8.1 Purpose of Simulation Studies

The simulation studies will evaluate both the physical behaviour of the aircraft model and the performance of the NMPC autopilot. The studies should answer whether the model can be trimmed, whether its responses are qualitatively reasonable and whether the controller can track references without violating actuator or flight-envelope constraints.

8.2 Planned Scenarios

The planned scenarios include:

1. altitude hold after an initial perturbation,
2. commanded altitude change,
3. commanded airspeed change,
4. heading hold and heading change,
5. simplified lateral path tracking,
6. descent along a prescribed vertical path,
7. localiser and glideslope capture,
8. asymmetric thrust or engine-response perturbation,
9. changed aircraft mass,
10. actuator saturation and rate-limit stress test.

8.3 Evaluation Metrics

The following metrics will be used to assess simulation results:

- steady-state tracking error,
- overshoot,
- settling time,
- maximum control-surface deflection,
- maximum control-surface rate,
- maximum thrust command,
- constraint violation magnitude and duration,
- smoothness of the response,
- numerical robustness of the simulation.

Chapter 9

Conclusions and Further Work

The thesis project establishes the foundations of a nonlinear flight dynamics simulation environment for a Boeing 777-like aircraft. The current work defines the coordinate conventions, state and input vectors, geometry data, mass and inertia estimates, standard atmosphere model, staged aerodynamic database, propulsion-load seed, control-surface actuator seed and validation-oriented reference cases. These elements provide the data layer and first force-and-moment adapters required before the full six-degree-of-freedom plant and NMPC controller can be integrated.

The most important modelling decision at the current stage is the separation of fixed aircraft properties from environment-dependent force quantities. Mass and inertia are stored as aircraft properties, while weight and thrust-to-weight quantities are computed using the current gravity assumption. Although the project presently uses fixed standard gravity, this interface leaves a clear path toward a future variable-gravity or higher-fidelity environment model.

Further work should focus on assembling the total force and moment build-up, integrating the nonlinear equations of motion, extending trim routines to the full coupled plant and validating the open-loop dynamic behaviour. After these steps, the longitudinal and lateral-directional NMPC controllers can be implemented and tested in the planned simulation scenarios.

The final engineering objective is not to reproduce a certified Boeing 777 simulator, but to obtain a transparent, consistent and extensible platform for studying flight dynamics and constrained autopilot design.

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